

Scalability & Future Plan

Date	02 July 2026
Team ID	XXXXXX
Project Name	Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application is hosted only on a local Flask server.	Cannot be accessed by remote users.	High
2	Uses a Linear Regression model only.	Prediction accuracy may be improved with advanced models.	Medium
3	No user authentication or database support.	User history and secure access are not available.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a single local user.	Deploy on Render, AWS, or Azure to support multiple users.
2	Data Storage	Uses a local CSV dataset.	Store data in MySQL or MongoDB for better management.
3	Performance	Uses a local CSV dataset.	Optimize code and use advanced machine learning models such as Random Forest or XGBoost.
4	Security	No login or authentication	Implement user authentication, role-based access, and secure data handling.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on a cloud platform (Render/AWS).	1 Month	Public accessibility and easier project sharing.
Phase 2	Integrate advanced machine learning models (Random Forest/XGBoost).	2 Months	Improved prediction accuracy and reliability.
Phase 3	Add user authentication and prediction history.	3 Months	Enhanced security and personalized user experience.
Phase 4	Integrate real-time HDI datasets and country comparison dashboard.	4 Months	More comprehensive analysis and better decision support.